

Module 08 [Advanced Topics: Java Programming]: (for Week 11:28-1/3/2026)

Topic 01 [java Programming: Anonymous object]

Problem Statement: Write a Java program that creates a Point object using anonymous object using the following getPoint() method.

```
Point pt=getPoint();
```

Code:

```
class Point
{
    int x;
    int y;
    Point(int x, int y)
    {
        this.x=x;
        this.y=y;
    }
    public static Point getPoint(int x, int y)
    {
        return new Point(x, y);
    }
    public void display()
    {
        System.out.println("Point("+x+", "+y+"");
    }
}

public class Topic01 {
    public static void main(String[] args) {
        Point pt = Point.getPoint(5,10);
        pt.display();
    }
}
```

Topic 02 [java Programming: ArrayList<> object]

Problem Statement: Write a Java program performs the following operations using ArrayList Object

- i) Declare a ArrayList object ax to store flowers
- ii) Add three flowers: Rose, Lilie, Carnation, Daisie, and Tulip

- iii) Insert a new flower Peonie in the second position of ax
- iv) Delete the flower Lilie
- v) Find any flower in ax
- vi) Sort the list
- vii) Delete the list

Code:

```
import java.util.*;
public class Topic2
{
    public static void main(String[] args)
    {
        ArrayList<String> ax = new ArrayList<>();
        ax.add("Rose");
        ax.add("Lilie");
        ax.add("Carnation");
        ax.add("Daisie");
        ax.add("Tulip");

        System.out.println("Initial Array: "+ax);

        ax.add(1,"Peonie");
        System.out.println("After Insertion: "+ ax);

        ax.remove("Lilie");
        System.out.println("After Deletion: "+ ax);

        String search = "Tulip";
        if(ax.contains(search))
        {
            System.out.println(search+"found in the list");
        }
        else
        {
            System.out.println(search + "not found");
        }

        Collections.sort(ax);
        System.out.println("Sorted list"+ ax);

        ax.clear();
        System.out.println("After Clearing: "+ ax);
    }
}
```

Topic 03 [inheritance: use of super keyword]

Problem Statement: For the following Java program write

- i) to display x of class A in class B
- ii) statement to call getX() of class A in class B
- iii) to call parameterized constructor of in class B

<pre>class A{ int x; public A(){ x=0; } public A(int x){ this.x=x; } public int getX(){ return(x+10); } class B extends A{ int x=20; public int getX(){ return(x+10); } } public class First{ public static void main(String[] args) {</pre>	
<pre> } }</pre>	

Code:

```
class A
{
    int x;
    public A()
    {
        x=0;
    }
    public A(int x)
    {
        this.x=x;
    }
    public int getX()
    {
        return(x+10);
    }
}
```

```

}
class B extends A
{
    int x=20;
    public int getX()
    {
        return(x+10);
    }
    public int getXa()
    {
        return super.x;
    }

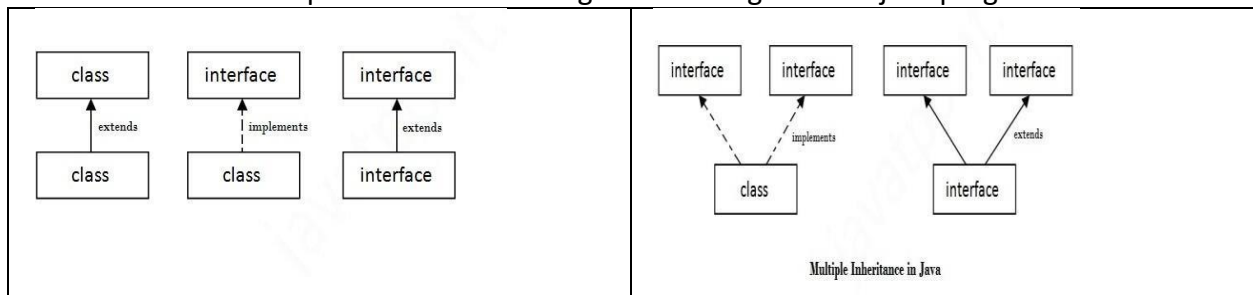
    }
    public void show()
    {
        System.out.println(super.getX());
    }
    B(int x)
    {
        super(x);
        System.out.println("Super Parameterized Constructor");
    }
}
}
public class First
{
    public static void main(String[] args)
    {

        B b = new B(10);
        System.out.println(b.getXa());
        System.out.println(b.getX());
        b.show();
    }
}
}

```

Topic 04 [inheritance]

Problem Statement: Implement the following relation diagram into java program



Code:

```
class A
{
    A()
    {
        System.out.println("A created");
    }
}

class B extends A
{
    //super(A);
}

public class Topic41 {
    public static void main(String[] args)
    {
        A b = new B();
    }
}

interface A
{
    void show();
}

class B implements A
{
    @Override
    public void show()
    {
        System.out.println("Implemnetd from A");
    }
}

public class Topic42 {
    public static void main(String[] args) {
        B a = new B();
        a.show();
    }
}

interface a
{

```

```

        void show();
    }
    interface b extends a {
        @Override
        default void show()
        {
            System.out.println("extened from a");
        }
    }
}
public class Topic43 {
    public static void main(String[] args) {
        b obj = new b() {};
        obj.show();
    }
}

interface A{
    void show();
}
interface B{
    void display();
}
class C implements A,B
{
    @Override
    public void show()
    {
        System.out.println("From a");
    }

    @Override
    public void display()
    {
        System.out.println("From b");
    }
}
public class topic44 {
    public static void main(String[] args)
    {
        C c = new C();
        c.show();
        c.display();
    }
}

```

```

interface A
{
    void showA();
}
interface B
{
    void showB();
}
interface C extends A,B
{
    @Override
    default void showA()
    {
        System.out.println("From A");
    }
    @Override
    default void showB()
    {
        System.out.println("From B");
    }
}
public class topic45 {
    public static void main(String[] args)
    {
        C c = new C(){};
        c.showA();
        c.showB();
    }
}

```

Topic 05 [Singleton class]

Problem Statement: Write a Java program to create and operate a singleton class.

```

class Singleton
{
    public static int count=0;
    private static Singleton[] instance= new Singleton[2];
    private Singleton()
    {
        count++;
    }
}

```

```

public static Singleton getInstance()
{
    if(count < 2)
    {
        instance[count] = new Singleton();
    }
    return instance[count];
}
public void show()
{
    System.out.println("Singleton instance");
}
}
public class Topic5
{
    public static void main(String[] args)
    {
        Singleton single1 = Singleton.getInstance();
        Singleton single2 = Singleton.getInstance();

        System.out.println(Singleton.count);
        System.out.println(single2.count);
        single1.show();

    }
}

```

Topic 06 [Java Multithreading]

Problem Statement: Write a program to run three applications i) Word Processor ii) Music Player and iii) Real-time clock simultaneously. Use run() method of Thread class or implements runnable interface.

```

class Wordprocessor extends Thread
{
    public void run()
    {
        for(int i=0; i<10; i++)
        {
            System.out.println("Word processor is running");
        }
    }
}
class Musicplayer extends Thread
{
    public void run()
    {
        for(int i=0; i<10; i++)

```



```

        {
            System.out.println("Musicplayer is running");
        }
    }
}
class Realtimeclock extends Thread
{
    public void run()
    {
        for(int i=0; i<10; i++)
        {
            System.out.println("Timer is on");
        }
    }
}
public class Topic6 {
    public static void main(String[] args)
    {
        Wordprocessor word = new Wordprocessor();
        Musicplayer music = new Musicplayer();
        Realtimeclock timer = new Realtimeclock();
        word.start();
        music.start();
        timer.start();

    }
}

```

```

class words implements Runnable
{
    public void run()
    {
        for(int i=0; i<10; i++)
            System.out.println("Ms Word is running");
    }
}
class music implements Runnable
{
    public void run()
    {
        for(int i=0; i<10; i++)
            System.out.println("Music is playing");
    }
}
class timer implements Runnable

```

```
{
    public void run()
    {
        for(int i=0; i<10; i++)
            System.out.println("Time is ticking");
    }
}
public class Topic6Runnable
{
    public static void main(String[] args)
    {
        words w = new words();
        Thread w1= new Thread(w);
        music m = new music();
        Thread m1 = new Thread(m);
        timer t = new timer();
        Thread t1 = new Thread(t);
        w1.start();
        m1.start();
        t1.start();
    }
}
```